

YRGYZ, Kazakhstan

WMO: 355420

Lat: 48.617N

Lon: 61.267E

Elev: 114

StdP: 99.96

Time Zone: 5.00 (E05)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.4	-25.5	-30.5	0.2	-27.0	-28.5	0.3	-25.3	16.4	-12.8	15.3	-7.1	3.3	340	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	36.0	18.2	34.1	17.9	32.2	17.5	19.8	30.8	19.1	30.5	18.5	29.6	4.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.0	22.9	15.5	11.1	22.5	14.2	10.3	22.5	57.2	30.5	54.8	30.9	52.6	29.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.5	11.8	9.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-30.5	22.5	2.7	1.8	-32.4	23.8	-33.9	24.8	-35.4	25.8	-37.4	27.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-13.5	-14.0	-7.0	8.7	16.6	23.1	24.9	22.1	15.8	6.5	-2.8	-9.8
	DBStd	15.09	6.48	6.75	6.44	6.26	4.40	4.08	3.47	3.60	4.57	4.66	5.53	6.26
	HDD10.0	3163	730	672	526	99	4	0	0	0	7	127	384	614
	HDD18.3	5051	988	905	784	292	86	9	2	11	101	366	634	873
	CDD10.0	1698	0	0	0	59	208	393	462	376	181	19	0	0
	CDD18.3	544	0	0	0	2	32	152	204	128	25	0	0	0
	CDH23.3	6393	0	0	0	37	398	1866	2466	1348	276	2	0	0
	CDH26.7	2824	0	0	0	4	112	891	1214	534	70	0	0	0
(14) Wind	WSAvg	4.3	4.4	4.9	5.0	4.4	4.5	4.0	4.0	3.8	3.9	4.1	4.1	4.0
(15) Precipitation	PrecAvg	174	11	11	12	16	18	20	17	12	12	15	16	14
	PrecMax	314	34	27	44	90	72	100	160	51	77	50	41	34
	PrecMin	79	1	0	0	0	0	0	0	0	0	0	2	2
	PrecStd	51	7	8	10	17	18	17	27	12	14	12	11	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.2	11.3	26.6	31.4	38.8	39.4	36.1	31.2	22.9	8.5	2.3
		MCWB	-0.2	0.3	6.9	13.9	15.4	18.8	19.1	17.8	15.1	12.3	4.9	1.1
	2%	DB	-0.5	0.2	6.0	23.9	29.6	36.1	36.2	33.6	28.6	19.6	6.4	0.8
		MCWB	-1.1	-0.4	3.4	12.9	14.7	18.2	18.5	17.5	14.5	11.3	3.4	0.1
	5%	DB	-1.7	-1.2	3.2	21.3	27.6	33.6	34.5	31.7	26.7	16.9	4.9	0.0
		MCWB	-2.1	-1.8	1.4	12.1	14.1	17.7	18.2	17.1	13.9	9.4	2.9	-0.5
	10%	DB	-3.2	-3.6	1.4	18.6	25.4	31.6	32.7	29.8	24.3	14.3	3.6	-1.0
		MCWB	-3.7	-4.2	0.4	10.9	13.7	17.2	17.9	16.6	13.2	8.0	1.8	-1.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.6	7.5	15.6	17.2	20.8	20.8	20.3	17.9	13.7	6.1	1.4
		MCDB	0.7	1.0	10.4	23.3	24.1	32.3	31.1	29.5	25.4	21.9	7.3	1.9
	2%	WB	-1.0	-0.4	3.5	14.0	16.3	19.7	20.0	18.9	16.4	11.8	4.2	0.4
		MCDB	-0.7	0.1	5.6	21.3	25.2	30.9	31.1	28.6	24.2	17.9	5.6	0.8
	5%	WB	-2.1	-1.8	1.7	12.6	15.3	18.7	19.4	18.2	14.8	10.2	3.1	-0.5
		MCDB	-1.8	-1.3	3.0	20.1	24.7	30.1	31.2	28.3	24.5	15.7	4.6	0.0
	10%	WB	-3.6	-4.2	0.4	11.2	14.4	18.0	18.8	17.5	13.7	8.5	1.8	-1.5
		MCDB	-3.2	-3.6	1.4	18.2	23.5	29.4	30.7	27.3	23.5	13.7	3.3	-1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.4	8.1	11.0	12.5	12.9	12.4	12.2	12.6	10.9	7.0	7.4
		MCDBR	9.9	7.2	7.8	14.4	15.2	15.5	14.6	14.3	15.4	14.8	8.5	6.2
		MCWBR	9.2	6.2	5.7	6.2	6.2	4.7	4.6	5.3	6.9	8.2	6.5	6.0
	5% WB	MCDBR	9.2	6.2	7.4	13.0	12.9	13.2	12.7	12.5	13.4	13.2	7.5	6.1
		MCWBR	8.8	6.1	5.6	6.1	6.0	4.8	4.9	5.4	6.6	8.0	6.3	5.9
(40) Clear-Sky Solar Irradiance	taub		0.308	0.334	0.401	0.431	0.429	0.424	0.416	0.415	0.365	0.355	0.341	0.317
	taud		2.094	2.023	2.015	2.065	2.151	2.226	2.266	2.247	2.339	2.279	2.213	2.128
	Ebn at Noon		716	787	792	813	834	840	841	823	833	767	678	648
	Edh at Noon		88	119	144	152	145	135	128	125	103	92	79	75
(44) All-Sky Solar Radiation	RadAvg		1.49	2.70	3.83	4.98	6.46	6.90	6.61	5.83	4.49	2.68	1.33	0.95
	RadStd		0.26	0.40	0.41	0.47	0.45	0.38	0.31	0.35	0.23	0.24	0.14	0.17

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page